

EXPERIMENT-8

IMPLEMENTATION OF 3DES

AIM:

To implement Triple DES (3DES) encryption and decryption using three keys.

ALGORITHM

1. Read the plaintext.
2. Read three keys K1, K2, K3.
3. Encrypt the plaintext using K1.
4. Encrypt the result using K2.
5. Encrypt the result using K3.
6. Display the encrypted text in hexadecimal format.
7. Decrypt using K3, then K2, then K1.
8. Display the decrypted plaintext.

Encryption:

$C = E(K3, E(K2, E(K1, P)))$

Decryption:

$P = D(K1, D(K2, D(K3, C)))$

CODE:

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char text[100], k1[20], k2[20], k3[20];
6     int i, len, n1, n2, n3;
7     printf("Enter message: ");
8     scanf("%s", text);
9     printf("Enter key 1: ");
10    scanf("%s", k1);
11    printf("Enter key 2: ");
12    scanf("%s", k2);
13    printf("Enter key 3: ");
14    scanf("%s", k3);
15    len = strlen(text);
16    n1 = strlen(k1);
17    n2 = strlen(k2);
18    n3 = strlen(k3);
19    /* Encryption */
20    for(i=0; i<len; i++)
21        text[i] ^= k1[i%n1];
22    for(i=0; i<len; i++)
23        text[i] ^= k2[i%n2];
24    for(i=0; i<len; i++)
25        text[i] ^= k3[i%n3];
26    printf("Encrypted Text: ");
27    for(i=0; i<len; i++)
28        printf("%02X ", (unsigned char)text[i]);
```

OUTPUT:

```
Output
Enter message: HELLO
Enter key 1: KEY
Enter key 2: ABD
Enter key 3: XYA
Encrypted Text: 1A 1B 10 1E 11
Decrypted Text: HELLO

=== Code Execution Successful ===
```

RESULT:

Thus, the Triple DES (3DES) encryption and decryption process was successfully demonstrated using three keys.